

Active Directory in Large Networks

IT Infrastructure Management (W04IST-SI0826G)

Wojciech Thomas, PhD

Department of Applied Informatics
Faculty of Information and Communication Technology

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This week

After this class, you should be able to:

- explain why Active Directory replication is necessary and what data is replicated
- describe the difference between intra-site and inter-site replication
- define what an AD site is and configure site topology using **AD Sites and Services**
- identify the five FSMO roles and explain which operate at domain level and which at forest level
- transfer an Operations Master role gracefully and seize it when the role holder is unavailable

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Why this matters

- A branch office loses connectivity to the main site — users still need to log in
- A password change in Warsaw must reach the domain controller in Kraków within minutes
- A domain controller hosting a critical FSMO role fails — what happens next, and what must you do?
- Today: the physical layer of Active Directory that keeps it running across multiple locations

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Agenda

1 Active Directory Replication

2 Operations Masters

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Quick recall

Without looking at your notes:

- 1 What is a domain controller?
 - 2 What tool do you use to manage Group Policy objects?
 - 3 What is SYSVOL?
 - 4 What does SYSVOL contain?
- 1 A server running AD DS that authenticates users and stores directory data
 - 2 **Group Policy Management Console (GPMC)** – gpmc.msc
 - 3 Shared folder on each DC that contains domain-wide configuration files
 - 4 Group Policy Templates (GPTs), logon scripts, and other domain-wide files

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What is replicated?

- Active Directory database
 - Partitions: Schema, Configuration, Domain, Application
 - Location: C:\Windows\NTDS
- SYSVOL volume

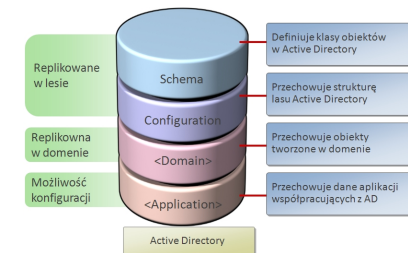


Figure 1: Active Directory database partitions

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Why replication matters

- The domain controls access to all resources
- No access to a domain controller = no access to resources
- Multiple DCs provide fault tolerance and local authentication in branch offices

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Before we continue – predict

- You have two domain controllers: one in Warsaw, one in Gdańsk
- A user changes their password in Warsaw at 9:00
- The Gdańsk DC has not yet received the update
- **What happens when the user tries to log in at the Gdańsk site at 9:02?**
- The Gdańsk DC forwards the authentication to the PDC Emulator (Warsaw) – the user logs in
- This is why the PDC Emulator role matters – it is the authoritative source for recent password changes

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AD Site

Site

A group of IP subnets connected by fast, reliable links (typically a LAN)

- Defined by the administrator:
 - Console: **AD Sites and Services** (dssite.msc)
 - Assign IP subnets to sites
- Sites represent the **physical** network topology; domains represent the **logical** structure

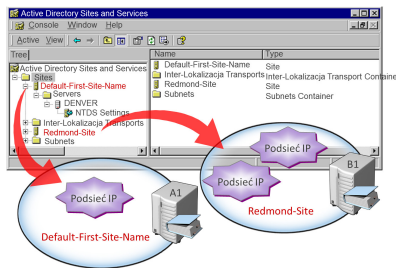


Figure 2: AD Sites

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Physical architecture of Active Directory

- Used to optimise replication traffic
- Components:
 - **Site** — subnet group
 - Domain Controller — hosts AD data for a site
 - Site Link — connection object between sites

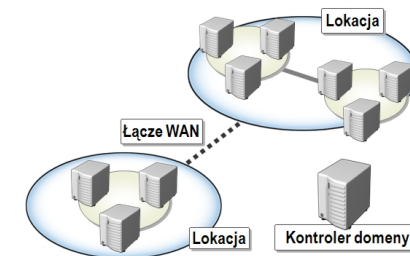


Figure 3: Active Directory physical architecture

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Intra-site replication — topology

- Ring topology with shortcut connections
- **KCC** (Knowledge Consistency Checker) service builds and maintains the topology automatically
- No admin configuration needed for topology within a site

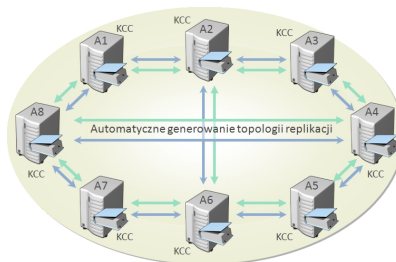


Figure 4: Intra-site replication topology

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Intra-site replication — characteristics

- Change-triggered (near real-time)
- Uses IP/RPC protocol
- Guarantees propagation within a site in under **5 minutes**

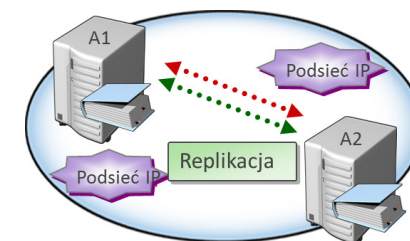


Figure 5: Intra-site replication

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Inter-site replication — topology

- Configured via **AD Sites and Services** console
- Administrator configures:
 - Assignment of domain controllers to sites
 - Site link types and schedules

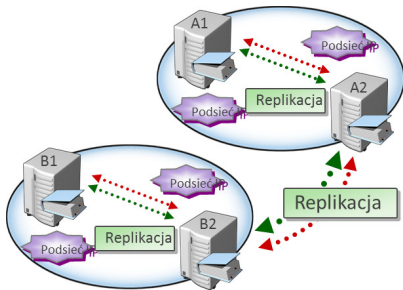


Figure 6: Inter-site replication topology

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Inter-site replication — characteristics

- Schedule-driven (not change-triggered by default)
 - Default interval: every **180 minutes**
- Supported protocols:
 - IP/RPC — preferred; requires a secure connection (VPN or leased line)
 - SMTP — requires certificate infrastructure; for asynchronous, non-domain partition data only

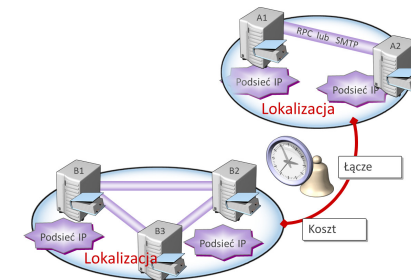


Figure 7: Inter-site replication

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Replication topology in a large domain

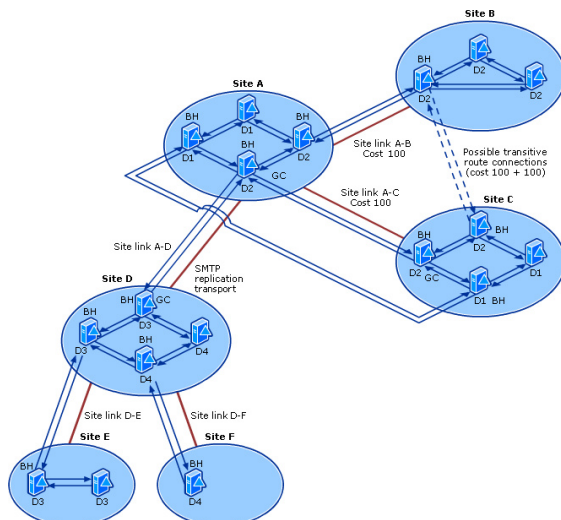


Figure 8: Replication topology across multiple sites

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Admin scenario

Your company has sites in Warsaw, Kraków, and Wrocław. Users in Wrocław report slow login times in the morning. The site link interval is set to 180 minutes. **As the admin, what are your three diagnostic questions?**

- 1 Is there a DC in the Wrocław site? (Users may be authenticating across the WAN)
- 2 Is the site link interval appropriate for this WAN link?
- 3 Is the KCC generating an optimal replication topology? (repadmin /showrepl)

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Agenda

1 Active Directory Replication

2 Operations Masters

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Operations Masters

- Active Directory is a fully distributed service
- Some operations cannot be safely performed on multiple DCs simultaneously

Operations Master (FSMO role)

A service running on **exactly one** domain controller

- Also known as **FSMO** – Flexible Single Master Operations

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Domain-level FSMO roles

In each domain, the following roles operate:

- **PDC Emulator** – first to learn of password changes; synchronises time for all DCs in the domain
- **RID Master** – issues pools of Relative Identifiers (RIDs) used to build unique SIDs for new objects
- **Infrastructure Master** – tracks object identifiers for objects moved from other domains

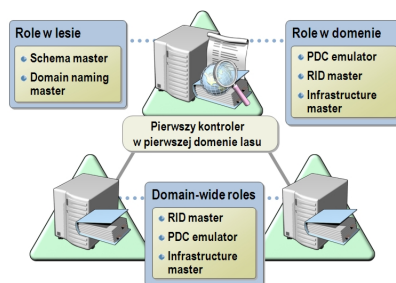


Figure 9: FSMO roles

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Forest-level FSMO roles

At the forest level, the following roles operate:

- **Domain Naming Master** – adds or removes domains in the forest
 - Console: **AD Domains and Trusts**
- **Schema Master** – processes all changes to the AD schema
 - Console: **Active Directory Schema snap-in (schmmgmt.dll)**

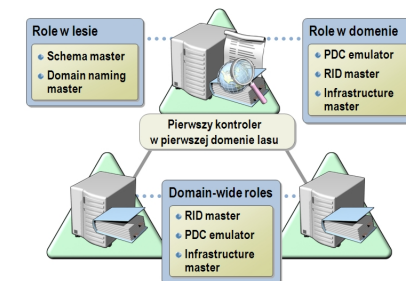


Figure 10: FSMO roles

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Checking FSMO role holders

```
1 # All five roles for the current domain and forest
2 netdom query fsmo

3 # Or via PowerShell (AD module)
4 Get-ADDomain | Select-Object PDCEmulator, RIDMaster,
   ↳ InfrastructureMaster
5 Get-ADForest | Select-Object DomainNamingMaster, SchemaMaster
```

↩

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Transferring FSMO roles

- Use when the current role holder will be decommissioned gracefully
- Performed via management consoles or PowerShell
- The role is "pulled" to the new DC while the old one is still online

```
1 # Transfer PDC Emulator to DC02
2 Move-ADDirectoryServerOperationMasterRole -Identity DC02
   ↳ -OperationMasterRole PDCEmulator
```

↩

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Seizing FSMO roles

- Use when the role holder has **failed** and cannot be recovered
- Force-transfer the role to another DC

Warning

After seizing a role, **never bring the old role holder back online**. It still believes it holds the role, which causes conflicts.

```
1 # Seize PDC Emulator (role holder unreachable)
2 Move-ADDirectoryServerOperationMasterRole -Identity DC02
   ↳ -OperationMasterRole PDCEmulator -Force
```

↩

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Before you go

On a piece of paper (or in your notes), write:

- 1 One thing from today you are confident about
- 2 One thing you are still unsure about

- Bring your question to the lab

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Bibliography



Hochschule
Bochum
University of Applied Sciences

- [1] A. Warren, *Exam ref 70-742 identity with windows server 2016*. Microsoft Press, 2017.
- [2] J. Krause, *Mastering windows server 2025 - fifth edition*. Packt Publishing, 2025.